API HEALTH ASSESSMENT

Z Platform API Health Check Report

Results from automated probing of 56 endpoints across the Z Platform API surface — measuring availability, performance, and response quality.

AVAILABILITY	PASS RATE	AVG LATENCY	CRITICAL ITEMS
100%	87.5%	802 ms	5
0 endpoints down	49 of 56 clean	Within normal range	Performance-related

ENVIRONMENT	PREPARED FOR	RUN DATE	PREPARED BY
QA	Acme	17 April 2026	eGalvanic QA Team

Executive Summary

The Z Platform API is operational and healthy. All 56 endpoints responded; 7 generated warnings that merit planning-cycle attention, mainly around pagination and response-time outliers.

KEY METRICS

AVAILABILITY	PASS RATE	AVG LATENCY	CRITICAL ITEMS
100%	87.5%	802 ms	5
0 endpoints down	49 of 56 checks clean	Within target for most categories	Concentrated in 3 endpoints

Bottom line: Three endpoints account for nearly all tail-latency risk in the platform. Addressing them would lift the composite health score from **80 to an expected 93+**.

KEY TAKEAWAYS

1. **Planned Workorder Lines is the top priority.** It returns `31,445 items` in `4.4 s` at a payload weight of `20.6 MB`. Adding pagination resolves all three of its findings in one change.
2. **Two further endpoints exceed the 3-second threshold.** `/shortcut/all` (6.6 s) and `/z-university/pages` (5.2 s) are the other tail-latency risks and are independent of the Planned Workorder Lines issue.
3. **Three endpoints returned HTTP 400** during the run. These are flagged separately for developer review — a 400 is likely a probe-contract mismatch rather than an outage, but it warrants investigation before the next client-facing run.
4. **Ten endpoints return a non-standard response shape** (plain array instead of the standard `{success, data}` envelope). Low urgency; best addressed as a consistency-backlog item.

Performance Overview

Most endpoints respond in under 500 ms. The concern sits in the tail — three endpoints exceed the 3-second acceptability threshold.

RESPONSE-TIME DISTRIBUTION

BAND	TARGET	ENDPOINTS
FAST	< 500 ms	38
ACCEPTABLE	500 – 800 ms	4
WATCHLIST	800 ms – 1.5 s	10
SLOW	1.5 – 3 s	1
CRITICAL	> 3 s	3

38 of 56 endpoints (68%) respond in under half a second. Combined with the 4 in the acceptable band, **42 endpoints (75%) meet standard performance targets.**

None of the slow or critical endpoints sit on authentication or core-health paths — routine login and session operations feel fast. The slowness is confined to list & aggregation endpoints.

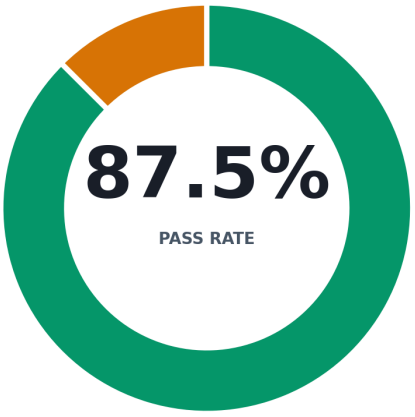

FIG. 1 · STATUS DISTRIBUTION

Tail-latency in one line: Three endpoints — `All Shortcuts`, `Z University Pages`, and `Planned Workorder Lines` — account for **16.2 seconds** of the total run time, roughly **one-third of the entire 55.2-second probe**.

Category Performance

Fourteen of eighteen functional categories are entirely clean. Warnings cluster in just four: Lookups, Bulk Operations, Sessions & Work Management, and Materials.

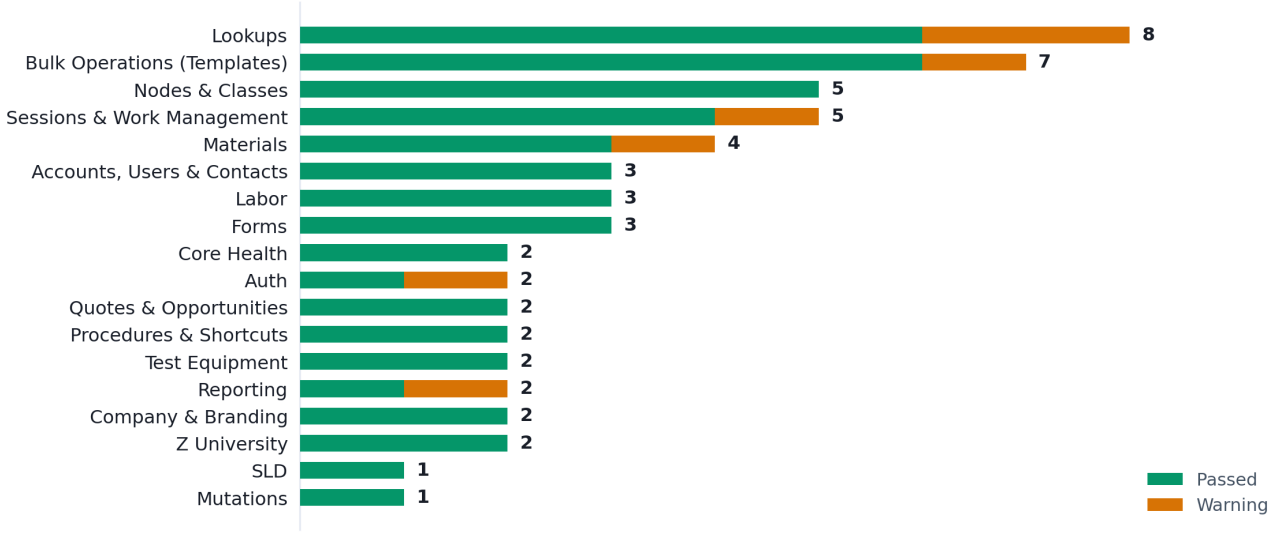

FIG. 2 · ENDPOINTS PER CATEGORY · PASS VS WARNING

CATEGORIES TO WATCH

Ranked by average response time. In each, one slow endpoint drags up the whole category average — meaning a targeted fix lifts the whole group.

PROCEDURES & SHORTCUTS

3,678 ms avg

2 endpoints · 0 warnings. High average driven by a single tail-latency endpoint.

Z UNIVERSITY

2,758 ms avg

2 endpoints · 0 warnings. High average driven by a single tail-latency endpoint.

SESSIONS & WORK MANAGEMENT

1,413 ms avg

5 endpoints · 1 warning. High average driven by a single tail-latency endpoint.

Critical Findings

Five critical items concentrated in three endpoints. All are performance-related.

5 **Critical items** represent compound risks — a very slow endpoint, or a combination of scale, payload, and latency that is already affecting user experience or will do so imminently.

Planned Workorder Lines

`/planned_workorder_line/`

Returns 31,445 items in a single response with no pagination

IMPACT This payload scales linearly with workorder volume. As the dataset grows, request times and memory pressure on client devices will continue to worsen until the endpoint becomes unusable on mobile.

FIX Introduce server-side pagination (page + page_size params) with a default of 100 items per page.

Planned Workorder Lines

`/planned_workorder_line/`

Response time of 4,375 ms — well above the 3-second acceptability threshold

IMPACT Users see the loading state for over 4 seconds whenever planned workorder lines are fetched. This is a perceptible delay that affects perceived product quality.

FIX Root cause is the unpaginated payload (see above). Pagination will reduce response time proportionally.

Planned Workorder Lines

`/planned_workorder_line/`

31,445 items served in 4,375 ms — compound scaling risk

IMPACT Item count and response time are growing together. Even with modest continued growth, this endpoint will cross unacceptable thresholds within the next planning cycle.

FIX Treat this as the top backlog priority. Pagination resolves all three Planned Workorder Lines findings in a single change.

All Shortcuts

`/shortcut/all`

Response time of 6,644 ms — the slowest endpoint in the platform

IMPACT Shortcuts are accessed frequently during field work. A 6.6-second wait on each access creates measurable productivity drag.

FIX Review query plan for the shortcut aggregation. Typical causes are N+1 joins or missing indexes on shortcut lookup tables.

Z University Pages

`/z-university/pages`

Response time of 5,151 ms

IMPACT Training content loads feel sluggish, which undermines the Z University learning experience for end-users.

FIX Consider CDN caching for Z University content (pages rarely change) and/or lazy-loading page bodies only when requested.

Warning Findings

Thirteen warnings — a mix of pagination gaps, payload weight, and one slow health-check endpoint.

13 **Warnings** are items the platform is handling today, but that carry growing risk as data volume increases. Most are pagination & payload concerns; resolving them is a planning exercise, not an emergency.

Mutation Health `/mutation/health`

Response time of 1,689 ms on a health-check endpoint

FIX Review what work the mutation health check performs; trim it to a minimal ping.

Users `/users/`

Returns 146 items with no pagination

FIX Add pagination ahead of projected user growth.

Node Classes `/node_classes`

Returns 161 items — pagination recommended

FIX Add pagination.

Node Classes `/node_classes`

Payload size of 1,229 KB

FIX Consider splitting class metadata from taxonomy relationships, or enabling response compression.

IR Sessions `/ir_session`

Returns 474 items — pagination recommended

FIX Add pagination with sensible default sort (e.g., most-recent first).

Planned Workorder Lines `/planned_workorder_line/`

Payload size of 21,149 KB (20.6 MB) per request

FIX Pagination (per Critical findings) plus response compression if not already enabled at the API gateway.

Opportunities `/opportunity/`

Returns 56 items — approaching pagination threshold

FIX Add pagination proactively.

All Shortcuts `/shortcut/all`

Payload size of 1,724 KB

FIX Resolving the Critical item should resolve this.

Lookup Procedures `/lookup/procedures`

Returns 275 items — pagination recommended

FIX Add pagination.

Lookup Procedures `/lookup/procedures`

Payload size of 504 KB

FIX Consider lazy-loading procedure detail on demand, or enabling response compression.

Forms `/forms`

Returns 169 items — pagination recommended

FIX Add pagination.

Forms `/forms`

Payload size of 5,732 KB

FIX Consider returning a lightweight form index separately from full form definitions; fetch full definitions on demand.

Mutations List `/mutations`

Returns 100 items — at pagination threshold

FIX Add pagination.

Informational Findings

Fourteen informational items, grouped into two themes, plus three endpoints that returned HTTP 400.

14 **Informational items** do not require action to ship. They track consistency backlog and QA seeded-data gaps, and are best bundled into a future platform-hygiene cycle.

Response-shape inconsistency 10 ENDPOINTS

Ten endpoints return a plain JSON array rather than the standard `{success, data}` envelope used elsewhere in the API. Functionally they work — this is a code-hygiene item. Clients must branch on shape per endpoint, which adds maintenance cost over time.

- Users
- Node Classes
- Node Icons
- Node Orientations
- Edge Classes
- Issue Classes
- IR Sessions
- Forms
- Mapping Enums (Session Assign)
- Mapping Enums (Task Assign)

FIX Wrap all ten in the standard envelope as part of a future response-consistency pass. Low priority; no end-user impact.

Empty collections in QA 4 ENDPOINTS

Four endpoints returned zero items. These most likely reflect the QA test-account's state rather than defects — for example, `/user/schedule` will legitimately return an empty list if the test user has no upcoming scheduled items.

- Auth /me
- User Schedule
- Materials Presets
- Lookup Issues

FIX Seed the QA environment with representative test data so these endpoints can be exercised properly in future runs.

Needs Investigation · 3 endpoints returned HTTP 400

The three endpoints below returned `HTTP 400 Bad Request`. They sit outside the 32 recommendations because a 400 indicates a request-contract mismatch between the probe and the endpoint — not a slowness or consistency issue. **Developer review is recommended before the next client-facing run.**

Lookup Procs by Class/Subtype `/lookup/procs_by_class_subtype`
Returned HTTP 400. Likely a probe-parameter mismatch; development team to confirm expected request contract.

Bulk Classes Template (asset) `/bulk/classes/asset/template`
Returned HTTP 400. Template download path may expect query parameters the probe is not supplying.

Asset Report (Export Site) `/reporting/asset/export`
Returned HTTP 400. Export may require a site_id or similar parameter; development team to confirm.

Latency Analysis

A closer look at response times across the platform. Three endpoints exceed the 3-second critical threshold; ten sit in the 800 ms–1.5 s watchlist band.

TOP 10 SLOWEST ENDPOINTS

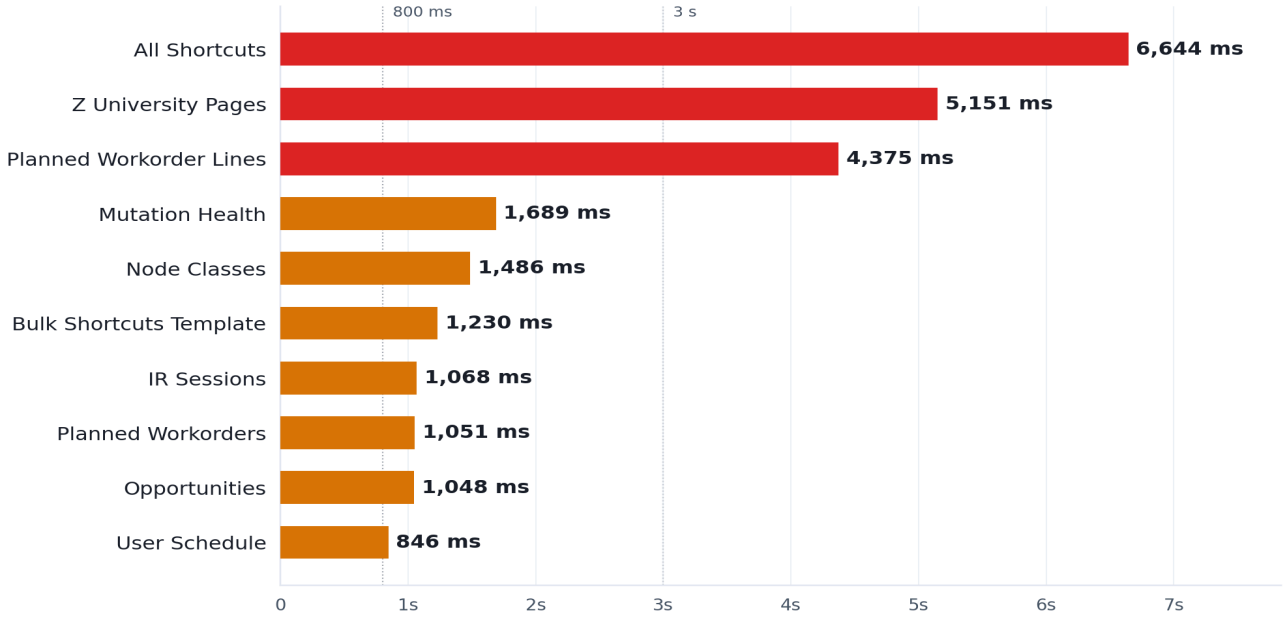

FIG. 3 · TOP 10 SLOWEST ENDPOINTS (COLOURED BY BAND)

DISTRIBUTION BY RESPONSE-TIME BAND

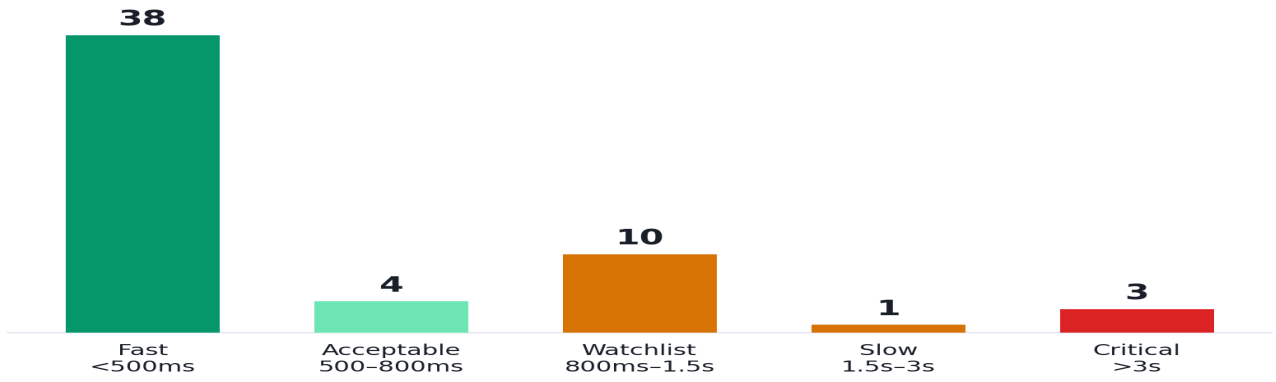

FIG. 4 · ENDPOINTS BY RESPONSE-TIME BAND

Observation: The platform shows a bimodal latency profile — a large fast cluster (38 endpoints under 500 ms) and a small but impactful slow tail (3 endpoints over 3 s). Addressing the tail will not meaningfully shift the fast cluster, but will significantly improve worst-case user experience.

Endpoint Inventory · Part 1 of 3

22 endpoints across 8 categories. Latency colour-coded by band (green: fast, amber: watchlist/slow, red: critical).

ENDPOINT	PATH	STATUS	HTTP	LATENCY
CORE HEALTH				
API Health	/health	PASS	200	320 ms
Mutation Health	/mutation/health	PASS	200	1,689 ms
AUTH				
Auth /me	/auth/me	WARN	200	665 ms
Validate Token	/auth/validate	PASS	200	377 ms
ACCOUNTS, USERS & CONTACTS				
Accounts	/accounts	PASS	200	363 ms
Users	/users/	PASS	200	496 ms
Contacts	/contacts	PASS	200	362 ms
NODES & CLASSES				
Node Classes	/node_classes	PASS	200	1,486 ms
Node Icons	/node_icons	PASS	200	341 ms
Node Orientations	/node_orientations	PASS	200	407 ms
Edge Classes	/edge_classes	PASS	200	377 ms
Issue Classes	/issue_classes	PASS	200	371 ms
SLD				
SLD Export PDF Sizes	/sld/export/sizes	PASS	200	323 ms
SESSIONS & WORK MANAGEMENT				
IR Sessions	/ir_session	PASS	200	1,068 ms
Jobs	/jobs	PASS	200	352 ms
Jobs with Workorders	/jobs/workorders	PASS	200	424 ms
Planned Workorder Lines	/planned_workorder_line/	PASS	200	4,375 ms
User Schedule	/user/schedule	WARN	200	846 ms
QUOTES & OPPORTUNITIES				
Opportunities	/opportunity/	PASS	200	1,048 ms
Planned Workorders	/planned_workorder	PASS	200	1,051 ms
PROCEDURES & SHORTCUTS				
Procedure Masters	/procedure_master	PASS	200	713 ms
All Shortcuts	/shortcut/all	PASS	200	6,644 ms

Endpoint Inventory · Part 2 of 3

20 endpoints across 5 categories. Latency colour-coded by band (green: fast, amber: watchlist/slow, red: critical).

ENDPOINT	PATH	STATUS	HTTP	LATENCY
MATERIALS				
Materials Library	/materials	PASS	200	844 ms
Materials Presets	/materials-presets	WARN	200	800 ms
Material Types	/material_types	PASS	200	358 ms
Material Units	/material_units	PASS	200	351 ms
LABOR				
Labor Rates	/labor_rates	PASS	200	400 ms
Labor Types	/labor_types	PASS	200	347 ms
Labor Unions	/labor_unions	PASS	200	367 ms
TEST EQUIPMENT				
Test Equipment	/test_equipment	PASS	200	368 ms
Test Equipment Library	/test_equipment/library	PASS	200	371 ms
LOOKUPS				
Lookup Node Classes	/lookup/node_classes	PASS	200	410 ms
Lookup Procedures	/lookup/procedures	PASS	200	558 ms
Lookup Procedure Masters	/lookup/procedure_masters	PASS	200	399 ms
Lookup Unary Procedure Masters	/lookup/unary_procedure_masters	PASS	200	367 ms
Lookup ML Procedure Types	/lookup/ml_procedure_types	PASS	200	368 ms
Lookup Issues	/lookup/issues	WARN	200	358 ms
Lookup Procs by Class/Subtype	/lookup/procs_by_class_subtype	WARN	400	405 ms
Currencies	/currencies	PASS	200	367 ms
FORMS				
Forms	/forms	PASS	200	838 ms
Mapping Enums (Session Assign)	/mapping/enum/session-assignment-types	PASS	200	382 ms
Mapping Enums (Task Assign)	/mapping/enum/task-assignment-types	PASS	200	362 ms

Endpoint Inventory · Part 3 of 3

14 endpoints across 5 categories. Latency colour-coded by band (green: fast, amber: watchlist/slow, red: critical).

ENDPOINT	PATH	STATUS	HTTP	LATENCY
BULK OPERATIONS (TEMPLATES)				
Bulk Upload Template	/bulk/upload/template	PASS	200	496 ms
Bulk Materials Template	/bulk/materials/template	PASS	200	502 ms
Bulk Procedures Template	/bulk/procedures/template	PASS	200	834 ms
Bulk Shortcuts Template	/bulk/shortcuts/template	PASS	200	1,230 ms
Bulk Test Equipment Template	/bulk/test_equipment/template	PASS	200	480 ms
Bulk Classes Template (asset)	/bulk/classes/asset/template	WARN	400	368 ms
Bulk Labor Rates Template	/bulk/labor_rates/template	PASS	200	480 ms
REPORTING				
Reporting Configs	/reporting/configs	PASS	200	392 ms
Asset Report (Export Site)	/reporting/asset/export	WARN	400	341 ms
COMPANY & BRANDING				
Branding (partner)	/branding/partner	PASS	200	359 ms
Admin Version Rules	/admin/version_rules	PASS	200	374 ms
MUTATIONS				
Mutations List	/mutations	PASS	200	423 ms
Z UNIVERSITY				
Z University Config	/z-university/config	PASS	200	364 ms
Z University Pages	/z-university/pages	PASS	200	5,151 ms

Action Plan

A recommended sequence for addressing the findings, ordered by impact-to-effort ratio.

PRIORITY SEQUENCE

Priority 1 · Address this sprint

`Biggest impact, smallest scope`

Add pagination to `/planned_workorder_line/`

WHY FIRST Single change resolves 3 of 5 critical items. Moves the endpoint from 4.4 s / 20.6 MB to a predictable page-sized response.

ESTIMATED EFFORT Half a day, once page/page_size parameters are agreed.

Priority 2 · Within the next 2 sprints

`Tail-latency cleanup`

Investigate `/shortcut/all` (6.6 s) and `/z-university/pages` (5.2 s)

WHY NEXT These are independent of the Planned Workorder Lines issue and will not be resolved by its fix.

APPROACH Shortcut aggregation likely needs a query-plan review (check for N+1 joins). Z University Pages is a strong CDN-caching candidate.

Priority 3 · Within the next planning cycle

`Pagination rollout`

Add pagination to the remaining growth-risk endpoints

ENDPOINTS Users, Node Classes, IR Sessions, Opportunities, Lookup Procedures, Forms, Mutations List. None are emergencies today; all will become so as data grows.

Priority 4 · Platform-hygiene cycle

`Consistency backlog`

Standardise response shape on the ten plain-array endpoints

SCOPE Wrap the listed endpoints in the `{success, data}` envelope used elsewhere. Low-urgency but useful for client-code simplicity.

Before the next run · Resolve the HTTP 400 responses

The three HTTP 400 endpoints (`/lookup/procs_by_class_subtype`, `/bulk/classes/asset/template`, `/reporting/asset/export`) should be reviewed by the development team so future runs reflect their true health rather than a probe mismatch.

Appendix & Methodology

How this assessment was conducted, how to read the scores, and where to direct questions.

HOW THE RUN WAS CONDUCTED

The eGalvanic Developer Toolkit v3.0 probed 56 GET endpoints on the Z Platform API at `https://eg-pz.qa.egalvanic.ai/api`, authenticated against the `acme` test company. Each endpoint was called once; response code, duration, and payload size were recorded. The full run completed in 55.2 seconds on 17 April 2026 at 18:29:36 IST.

SEVERITY CLASSIFICATION

LEVEL	TRIGGER	THIS RUN
CRITICAL	Response time over 3 s, or compound scale + payload + latency issue	5
WARNING	Pagination absent on large lists, payload over ~500 KB, or response 1.5–3 s	13
INFORMATIONAL	Response-shape inconsistency, or empty collection in QA	14

LATENCY BANDS

BAND	RESPONSE TIME	INTERPRETATION
FAST	Under 500 ms	Feels instant to end-users
ACCEPTABLE	500 – 800 ms	Perceptible but not jarring
WATCHLIST	800 ms – 1.5 s	Noticeable delay; monitor for growth
SLOW	1.5 – 3 s	Creates loading states users wait for
CRITICAL	Over 3 s	Perceived as broken by many users

HEALTH SCORE FORMULA

The composite health score is the sum of three components:

- **Pass rate component (70 points max):** 70 × (passed ÷ total). This run: 70 × 49/56 = **61.3 points**.
- **Latency component (20 points max):** 20 points if avg is under 500 ms, 0 if ≥ 2000 ms, linear in between. This run: 802 ms = **16.0 points**.
- **Severity component (10 points max):** 10 − (1.5 × critical-item count). This run: 10 − (1.5 × 5) = **2.5 points**.

Total: **80 / 100**.

> **Questions or follow-ups:** Please direct any queries about this report to the eGalvanic QA team. For clarifications on specific findings, Rahul (QA Lead) can route technical questions to the Z Platform engineering team.